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Gene-symbol queries to MyVariant.info drop ClinVar missense variants without warning, and a 3D label-proximity score shows no gain over a neighbour count

Aaryav Sharma[1,*]

[1]Eastview High School, Minnesota, USA

[*]Corresponding author. aaryavvatts@gmail.com

Abstract

Motivation: Studies of variant interpretation often begin by pulling a gene's ClinVar variants through an annotation service, and every later number inherits whatever that step loses. We know of no published measurement of how complete it is. We also asked whether a 3D score built from the positions of labelled pathogenic and benign residues (3D-PPS, and a shrunk version, S1) beats a plain count of labelled neighbours.
Results: A MyVariant.info query on the gene-symbol field raised no error but lost more than 5% of ClinVar missense variants in all 19 genes tested, and in 97 of the 100 genes with the most variants. About 70% of the loss was snapshot lag, because the service mirrors ClinVar from May 2025. About 30% came from the symbol field, which holds one symbol per record; that field alone cost more than 5% in five genes. A simple count against the ClinVar release finds both causes. In a ten-gene panel (*MYH7* plus nine non-cardiac genes), S1 was 0.134 AUROC below AlphaMissense and 0.032 below the neighbour count, and added at most 0.003 to a model that already had the count. This equivalence is weak, because the panel's predictors alone score 0.92 to 0.99.
Conclusions: For this service and query, retrieval completeness should be checked against the source release before any downstream analysis. In this panel, a label-proximity score gave no measurable gain over a neighbour count.

Keywords ClinVar, MyVariant.info, variants of uncertain significance, AlphaFold, data retrieval, reproducibility

Introduction

Variant classification leans on computational evidence (Richards et al., 2015; Pejaver et al., 2022), and studies of that evidence need a variant set. A common way to get one is to ask an annotation service for the ClinVar records of a gene (Landrum et al., 2018; Xin et al., 2016). If that step loses variants without an error, every later number inherits the loss. Gene-name errors and annotation choices are known to change genomic results (Ziemann et al., 2016; McCarthy et al., 2014), but we know of no published measurement of how many ClinVar missense variants a standard gene-symbol query drops.

Protein structure gives a second kind of evidence. Pathogenic variants cluster in 3D within genes (Homburger et al., 2016; Quinodoz et al., 2022; Iqbal et al., 2020), and AlphaFold models make such analyses possible for most proteins (Jumper et al., 2021; Varadi et al., 2022). A proximity score uses known labels at prediction time, so its measured accuracy depends on where the labels sit (Grimm et al., 2015; Rastogi et al., 2025). We ask whether a label-proximity score beats what coverage alone gives, a count of labelled neighbours.

We make two measurements. First, we measure retrieval loss in the ClinVar mirror inside MyVariant.info for 19 candidate genes and for the 100 genes with the most missense variants. Second, we evaluate a 3D Pathogenic Proximity Score (3D-PPS) and a shrunk version, S1, in a ten-gene panel chosen by a written inclusion rule. The index gene is *MYH7*, which encodes beta cardiac myosin and carries many variants of uncertain significance (VUS) (Friedman et al., 2024; Fowler and Rehm, 2024). The scope is narrow on purpose. We do not claim that proximity scores fail in general. We claim that, for this score family, ClinVar labels and this panel, the score showed no gain beyond a neighbour count, which is itself computed from the 3D structure.

Methods

Design and status of the analysis plan

A written analysis plan was prepared before any cross-gene result was computed. It fixed the gene panel, the inclusion rule, the benign sets, the scores, the primary estimand, the tests, the smallest difference of interest (0.02 AUROC) and the correction for multiple tests. It is a written plan, not a registration: it was not deposited in a registry, and its time stamps come from the project's own logs. It was written

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after the *MYH7* results of an earlier internal analysis were known, and the choice of S1 as the primary score was motivated by those results. The *MYH7* analyses are therefore confirmatory only in the sense that estimands, margins and tests were set before they were rerun. After work began we enlarged the gene pool (see Gene panel and inclusion rule), logged six deviations once the cross-gene job was already printing per-gene results, and added two exploratory analyses: the audit of the 100 largest genes and the union-set contrasts between benign sets. Verdicts are reported as verdicts under a written plan.

Data and software

We used the ClinVar file variant_summary.txt.gz (server date 5 October 2026, retrieved 7 October 2026 UTC; SHA-256 in the repository), kept single-nucleotide variants whose HGVS name gives a missense change, paired GRCh37 and GRCh38 rows by VariationID, and checked the reference amino acid against the protein model. Allele frequencies come from gnomAD v4 through its GraphQL service (gnomad_r4, GRCh38; Guez et al., 2026, preprint). AlphaMissense (Cheng et al., 2023), REVEL (Ioannidis et al., 2016) and 13 further predictor scores come from dbNSFP 4.8a (Liu et al., 2020), read through MyVariant.info (build of 24 June 2025, ClinVar source version 2025-05). Structures are AlphaFold DB model version 6 files for the UniProt canonical accessions (UniProt Consortium, 2023). We also used the ClinGen Evidence Repository table (Rehm et al., 2015). Software was Python 3.9.6 with NumPy, SciPy, scikit-learn, statsmodels, pandas and Biopython (Harris et al., 2020; Virtanen et al., 2020; Pedregosa et al., 2011; Cock et al., 2009). All seeds are 42.

Retrieval audit

For each gene we ran the query `clinvar.gene.symbol:GENE` against MyVariant.info, collected all result pages, and compared the ClinVar VariationIDs returned with the missense SNVs in the bulk file. Each missing variant was looked up again by its GRCh37 coordinate and put in one of three classes: a record with a different gene symbol (symbol-field loss); no ClinVar record at all in the service (absent from the snapshot, which we call snapshot lag); or no GRCh37 coordinate. The first two are reported separately, because snapshot lag is a property of the service's release date and symbol-field loss is a property of its data model. We report loss per gene and the number of genes with more than 5% loss, with an exact binomial interval. The gene sets were the 19 candidates of the panel section (part of the plan) and the 100 genes with the most ClinVar missense SNVs (added during the work; exploratory).

Labels and benign sets

The unit is the protein change. It is pathogenic (P) if some ClinVar record is Pathogenic or Likely pathogenic and none is benign. It is benign (B) if some record is Benign or Likely benign and none is pathogenic, uncertain or conflicting; we use this strict rule because a benign call is a claim about every allele that codes for the change. Other uncertain or conflicting changes are VUS and are not scored. Three benign sets are used. T1 (curated) keeps benign changes with a review status of two or more stars or a benign assertion in the ClinGen Evidence Repository. T2 (primary) keeps any ClinVar benign change, and T1 is a subset. T3 (*MYH7* only) adds changes with no ClinVar record whose popmax filtering allele frequency is at least 2×10^{-4}, the BS1 value of Kelly et al. (2018), with an allele count of at least 5 in the group that gives the maximum. For continuity we also re-ran two global-frequency stand-in sets (any frequency of at least 10^{-5} or 5×10^{-5}). These lines lie far below the BS1 value, so they serve only as comparison sets.

The scores

For a variant at residue r, let $d_P(r)$ and $d_B(r)$ be kernel-weighted densities of pathogenic and benign labelled residues in its neighbourhood, using only labels of other residues:

$$d_X(r) = \sum_{q \in X} \exp\left(-\frac{\delta_{rq}^2}{2\sigma^2}\right) \frac{l_q}{100}, \tag{1}$$

summed over residues q with C-alpha distance $\delta_{rq} \le 10$ Å, more than 4 apart in sequence, and AlphaFold confidence $l_q \ge 50$. The width is $\sigma = 5$ Å. These settings were fixed in the plan and not tuned. S0 is the original score, $(d_P - d_B)/(d_P + d_B)$, undefined when the denominator is 0. S1, the primary score, is $(d_P - d_B)/(d_P + d_B + k)$ with $k = 1$; it is defined for every variant and breaks ties. We chose S1 and $k = 1$ after seeing the earlier *MYH7* results, so our claims concern that choice and not every shrunk score. S2 is a class-balanced version. The comparators are the neighbour count (labelled residues in the 10 Å shell and more than 4 apart in sequence), a 1D version of S1 that uses sequence distance, AlphaMissense and REVEL. AUROC counts ties as one half, and the tie fraction is reported with every AUROC. AlphaMissense and REVEL are partly trained on clinical or frequency data, so they are upper-bound yardsticks and not independent benchmarks.

Gene panel and inclusion rule

The candidate panel had 19 genes: eight sarcomere genes (*MYH7, MYBPC3, TNNT2, TNNI3, TPM1, ACTC1, MYL2, MYL3*), *LDLR, FBN1, SCN5A, KCNQ1, KCNH2, PTEN, MLH1, MSH2, MSH6, BRCA1* and *TP53*. A gene was evaluated if it had at least 100 pathogenic and 20 benign missense changes in T2 (I1), one full-length AlphaFold DB model (I2), a reference amino acid that matched the model in at least 95% of labelled variants (I3), and AlphaMissense and REVEL scores for at least 80% of them (I4). *MYH7* is the index gene and is always reported. Only six of the 19 genes passed. Amendment A1 then enlarged the pool to the 100 genes with the most ClinVar missense SNVs, with the stated purpose of reaching at least 10 evaluable genes. It was written after the label counts of the 19 genes were known and before any performance result was computed. The rule is applied the same way to every gene, but the decision to enlarge the pool was made because the first panel was small. It added *TSC2, SCN1A, CACNA1A* and *ABCA4*, giving ten genes. The rule favours genes with many benign missense variants, which are mostly not cardiac genes, and excludes very large proteins that have no single full-length model (10 of the 20 pool genes that passed I1 failed I2).

Inference

The primary estimand is the AUROC of S1 for P against B changes in T2. The primary null shuffles class labels within strata of neighbour-count bin and burial tertile and recomputes the score (10,000 shuffles, one-sided add-one p-value); the unconditional null shuffles all labels and cannot separate class structure from coverage. Intervals come from a residue-cluster bootstrap, with a delete-one-benign-cluster

jackknife for genes with about 20 benign changes. Power was simulated under the observed tie structure. Nested logistic models use the 15 predictor scores plus burial and confidence (the panel), plus the neighbour count, plus S1, or both, with 20 repeats of residue-grouped five-fold cross-validation and structural features recomputed from training labels in every fold. For every contrast we report the estimate, its interval, the minimum detectable difference and a bootstrap test of equivalence at 0.02; if the minimum detectable difference exceeds 0.02 the verdict is *underpowered*. P-values are Holm-corrected within six families listed in the plan. Per-gene differences were pooled by random-effects meta-analysis with the Hartung-Knapp interval (DerSimonian and Laird, 1986; IntHout et al., 2014).

Reproducibility and ethics

One script rebuilds all tables, figures and the paper from archived raw downloads. Live services change over time, so reruns reuse the archived downloads, and we did not test whether a fresh download reproduces our data. No human subjects or identifiable data were involved.

Results

A gene-symbol query lost variants in every gene tested

The query lost more than 5% of ClinVar missense variants in all 19 candidate genes and returned no error or warning in any of them (Table 1, Figure 1). In the exploratory audit, 97 of the 100 genes with the most variants lost more than 5% as well; the median loss was 11.4% for the candidates and 15.8% for the top 100, and 32 of the top 100 lost more than 20%. Two mechanisms explain the loss, in the same proportions in both sets. About 70% of lost variants are absent from the service's ClinVar snapshot (May 2025), whereas our file is from October 2026; this is snapshot lag. About 30% have a record in the service under a different gene symbol, because the service stores one symbol per record.

The two mechanisms matter differently. Snapshot lag alone exceeded 5% in every candidate gene (smallest 5.7% in *BRCA1*); in *BRCA1, TP53, MSH2, MLH1* and *MSH6* the 5% line is crossed only because of lag. A fresher mirror would shrink this part. The symbol-field loss is structural and gene-specific: it alone exceeded 5% in five genes (*ACTC1, MYL2, MYH7, KCNQ1, SCN5A*) and was zero in ten. In *ACTC1* it affects 317 of 370 variants. In *MYH7* the lost records carry the symbols LOC126861897 (299 variants), LOC126861898 (238) and MIR208B (1), which are regulatory-region and microRNA records inside the gene (van Rooij et al., 2009). In four candidate genes, and in eight of the top 100, a run of residue bins lost 90% or more of its variants (in *MYH7*, residues 1548 to 1740). The loss also falls on labelled variants, more heavily on benign ones: in the 19 genes 10.1% of pathogenic (467 of 4,630) and 18.5% of benign (235 of 1,268) nucleotide-level records were not returned. A count of retrieved against expected variants catches both mechanisms. We use "without warning" only in the sense that no error or warning was raised.

Ten genes meet the inclusion rule, and the other sarcomere genes do not

Ten genes passed (Table 2): *MYH7, LDLR, MLH1, MSH2, BRCA1, TP53* and, from the enlarged pool, *TSC2, SCN1A, CACNA1A* and *ABCA4*. Apart from *MYH7*, none is a cardiomyopathy gene. The other sarcomere genes failed because ClinVar has almost no benign missense variants for them (*TNNT2* 4, *TNNI3* 2, *TPM1* 3, *ACTC1* 1, *MYL2* and *MYL3* none), and *MYBPC3* has 35 benign but only 33 pathogenic changes. *FBN1* has no AlphaFold DB model, and *SCN5A* has a numbering mismatch between the ClinVar transcript and the UniProt sequence. Our conclusions about cardiology therefore rest on *MYH7* alone, with 20 benign changes.

In *MYH7* the original score is capped, and the estimand matters

In *MYH7*, 7 of the 20 benign variants have a labelled neighbour. All 7 score exactly 1.0 on S0, and so do most pathogenic variants: 96.2% of P–B pairs are tied, none favour pathogenic and 3.8% favour benign. The AUROC of S0 on scoreable variants is therefore 0.481 and cannot exceed 0.5 on this sample (Figure 2A); its narrow interval reflects ties, not precision. The result changes with the handling of undefined scores: S0 with undefined set to neutral gives 0.693 (95% CI 0.573 to 0.804) on all 399 variants, S1 gives 0.760 (0.660 to 0.840) and the class-balanced S2 gives 0.742. The neighbour count gives 0.775 (0.683 to 0.849), the 1D version of S1 gives 0.794, and AlphaMissense and REVEL give 0.914 and 0.899. The tie fraction falls from 0.96 for S0 to 0.16 for S1. The pattern holds with the curated set T1 (12 benign) and with the global-frequency stand-ins (Figure 2B,C).

On the same variants, S1 minus the neighbour count is −0.016 (−0.041 to +0.008; minimum detectable difference 0.035), and S1 minus AlphaMissense is −0.154 (−0.275 to −0.040). The first comparison is underpowered and cannot show equivalence. Benign variants are also less often near a labelled residue than pathogenic ones: 65.0% of benign (13 of 20) and 24.3% of pathogenic variants have no labelled neighbour (exact odds ratio for having a neighbour 5.76, 95% CI 2.06 to 17.60). The neighbour count alone separates the classes with AUROC 0.775. All 12 curated benign calls in *MYH7* come from the ClinGen cardiomyopathy expert panel and cite the frequency criteria BA1 or BS1, so every benign set here is frequency-defined. On variants shared between T2 and T1, the AUROC of S1 changed by only −0.003.

Across ten genes S1 is below both AlphaMissense and the neighbour count

S1 had a lower AUROC than AlphaMissense and than REVEL in all ten genes (Figure 3), and was below the neighbour count in nine of ten, with the interval below zero in four. In the random-effects analysis S1 minus AlphaMissense was −0.134 (Hartung-Knapp 95% CI −0.180 to −0.088; I^2 81%; prediction interval −0.278 to +0.009; Holm-adjusted $p = 0.0002$), and S1 minus the neighbour count was −0.032 (−0.059 to −0.005; I^2 67%; prediction interval −0.095 to +0.031; adjusted $p = 0.026$). The plan classed the second result as unfavourable to S1. The neighbour count is itself 3D-derived, so this compares two label-based 3D summaries. Results are similar without *MYH7* (−0.133 and −0.037). A 1D version of S1 did not differ from S1 on average (pooled S1 minus 1D +0.022, −0.038 to +0.083; exploratory), but the sign varies by gene, so the pooled

Table 1 Retrieval loss of the single gene-symbol query. Loss is the share of ClinVar missense SNVs of a gene (October 2026 file) that the query did not return.

Gene set	Genes losing > 5% (95% CI)	Median loss (%)	Pooled loss (%)	Other symbol stored	Absent from snapshot
19 candidate genes	19 of 19 (0.82 to 1.00)	11.4	12.9 (5,042 of 39,174)	1,501 (29.8%)	3,531 (70.0%)
100 genes with most ClinVar missense SNVs	97 of 100 (0.91 to 0.99)	15.8	19.5 (61,510 of 315,128)	18,407 (29.9%)	43,060 (70.0%)

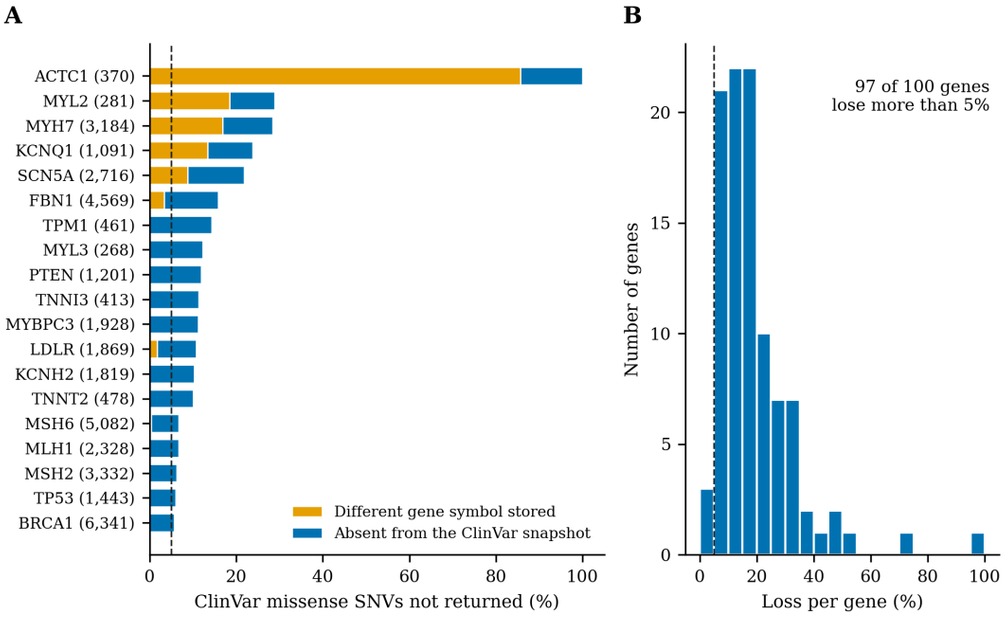


Figure 1 Retrieval loss by gene. (A) The 19 candidate genes, split by mechanism; the number of ClinVar missense SNVs is in brackets and the dashed line marks 5%. (B) The 100 genes with the most ClinVar missense SNVs (exploratory). Variants with no GRCh37 coordinate (10 of 5,042 lost in A) are not drawn.

Table 2 The ten evaluable genes (benign set T2). P and B are pathogenic and benign changes; S0 ties is the share of P–B pairs tied among scoreable variants; AM is AlphaMissense.

Gene	P	B	S0 ties	S1	Count	AM
MYH7	379	20	0.96	0.76	0.78	0.91
ABCA4	631	26	0.90	0.72	0.77	0.97
MLH1	186	35	0.59	0.81	0.86	0.93
LDLR	699	37	0.52	0.88	0.86	0.98
SCN1A	1023	59	0.84	0.92	0.92	0.97
CACNA1A	163	102	0.46	0.85	0.89	0.97
TP53	232	144	0.03	0.88	0.93	0.98
TSC2	185	177	0.12	0.71	0.81	0.94
MSH2	142	265	0.07	0.78	0.79	0.93
BRCA1	215	424	0.05	0.77	0.89	0.94

test does not rule out a 3D benefit in some genes. Across genes the AUROC of the neighbour count rose with the coverage asymmetry between classes (Spearman $\rho = +0.81$, $p = 0.005$), while that of S1 did not ($\rho = +0.35$, $p = 0.33$); these are two of about 28 uncorrected across-gene analyses and are exploratory.

Nested models and the coverage-and-burial null

Adding S1 to a model that already had the predictor panel and the neighbour count changed the cross-validated AUROC by between -0.002 and $+0.003$ across the ten genes (Figure 4B); the largest upper confidence limit was $+0.010$ (*MSH2*). Equivalence within 0.02 was therefore shown in all ten genes. This is weaker than it looks: the panel alone already reaches 0.92 to 0.99, so there is little room to add anything, S1 is nearly a monotone function of the neighbour count, and the bootstrap intervals hold out-of-fold predictions fixed and so understate variance. We read it as no measurable gain in this design, not as proof that 3D information is redundant. Against the coverage-and-burial null, S1 was supported in one of ten genes (*MSH2*: 0.780 against a null mean of 0.609; Holm-adjusted $p = 0.001$). In three genes the null was not rejected with adequate power, and in six it was not rejected but the test was underpowered. In *MYH7* S1 scored 0.760 against a null mean of 0.740 ($p = 0.166$; smallest detectable excess 0.051). Under the unconditional null S1 reached the resampling floor ($p = 0.0001$), which shows only that class is spatially structured and that coverage and burial explain most of it. Table 3 counts the verdicts of the six test families in the plan.

Discussion

What the results support. A single gene-symbol query lost variants in every gene we tested, and the loss is not small (median 11 to 16%). About 70% is snapshot lag, which a fresher mirror would reduce, and about 30% is the symbol field, which is structural

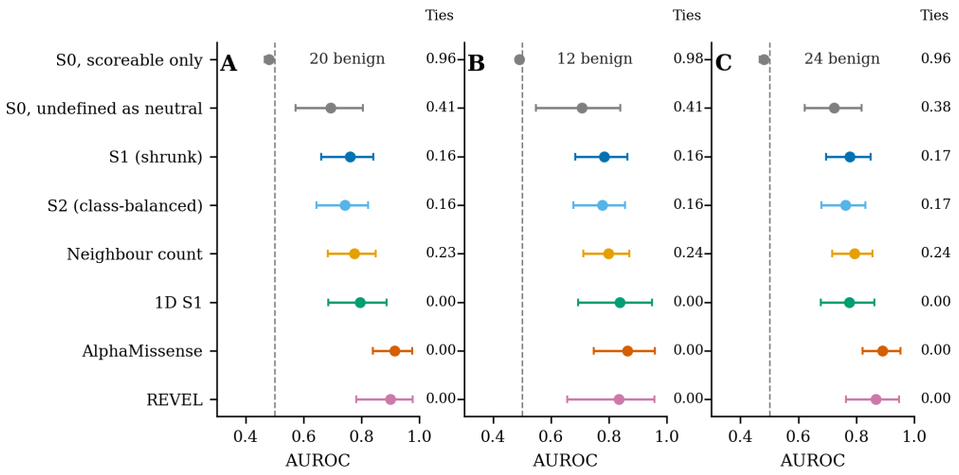


Figure 2 AUROC in *MYH7* for the original score and its alternatives, with 95% residue-cluster bootstrap intervals. (A) Benign set T2 (any ClinVar benign, 20 benign). (B) T1 (curated, 12 benign). (C) Global-frequency stand-ins at 5×10^{-5} or more (24 benign). The column of numbers is the tie fraction. The dashed line marks 0.5.

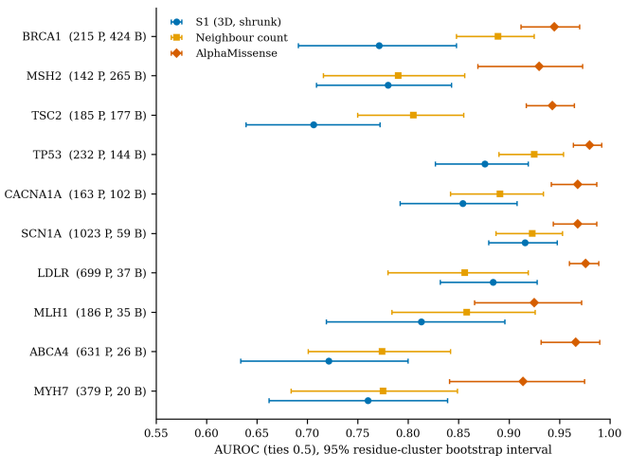


Figure 3 AUROC of S1, the neighbour count and AlphaMissense in the ten genes (T2), with 95% residue-cluster bootstrap intervals.

Table 3 Verdicts of the tests in the written plan, by family. F2 and F6 hold a superiority and an equivalence test per contrast, so most of their supported results are equivalence results. Sup., supported; Not sup., not supported; Under., underpowered; Unfav., unfavourable.

	Question	Sup.	Not sup.	Under.	Unfav.
F1	*MYH7*: score vs coverage null	0	6	10	0
F2	*MYH7*: nested models	8	10	6	0
F3	*MYH7*: paired contrasts	3	0	6	1
F4	Ten genes: S1 vs coverage null	1	3	6	0
F5	Ten genes: pooled S1 vs AM and count	1	0	0	1
F6	Ten genes: S1 beyond panel plus count	10	8	0	2

and large in a few genes. We did not test other annotation services or query fields, so the result concerns this service and this query. The checks (a count against the ClinVar release, and a snapshot date) apply to any route. For the score, once a count of labelled neighbours is available, S1 adds no gain we can measure, and a predictor that has seen other evidence is better. Because the count is itself 3D-derived, we have shown that a label-proximity score does not improve on a simpler 3D label summary, not that structure carries no information. A plausible reason why a proximity score looks informative is that benign variants lie far from labelled residues more often than pathogenic ones, as in *MYH7* and in every gene of the panel; the neighbour-count AUROC tracks this asymmetry across genes, but S1 does not, so we offer it as a plausible explanation for the count only. In *MYH7* what moved the AUROC most was the handling of undefined scores and ties (0.481, 0.693, 0.760), and the benign definition moved S1 by at most 0.003 on shared variants.

Reporting checks. Our results suggest five checks that were not in the reporting lists we know (Bossuyt et al., 2015; Collins et al., 2024; Walsh et al., 2021), or add detail to items that are: (1) compare the number of variants retrieved with an independent count, and compare the snapshot date or highest record identifier with the release you believe you are using; (2) give pathogenic and benign counts and the share of tied pairs beside every AUROC; (3) state how undefined scores are handled and report one score defined for all variants; (4) compare a label-based score with a plain count of labelled neighbours and test it against a null that preserves coverage and burial; (5) fix a smallest difference of interest in advance and report intervals and minimum detectable differences. We have not tested these checks on other studies.

Limitations. The panel is small and, apart from *MYH7*, not cardiac; four of its ten genes entered through an amendment written after we saw label counts. Large proteins without a full-length AlphaFold model are excluded and *SCN5A* was lost to a numbering mismatch. The plan was not registered externally, and it was written when the earlier *MYH7* results were known. The equivalence results are weak. ClinVar labels can reflect proximity reasoning themselves

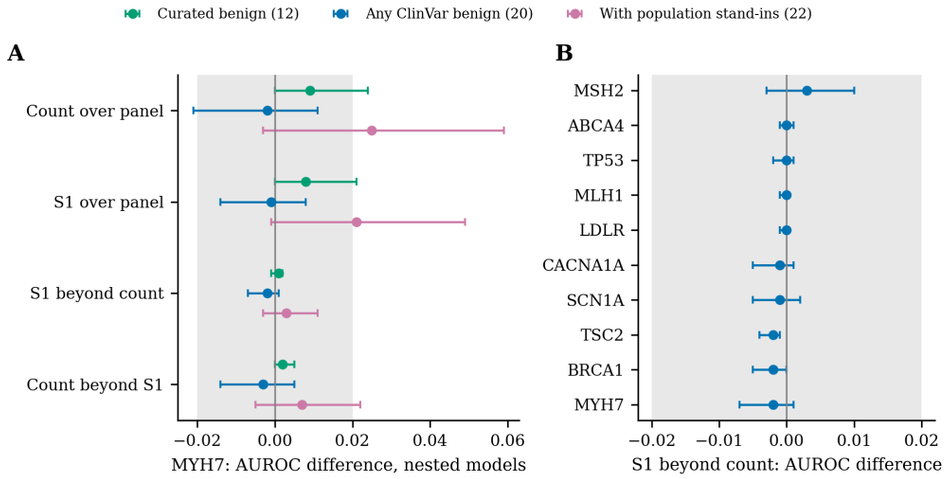


Figure 4 Nested-model AUROC differences from 20 repeats of residue-grouped five-fold cross-validation, with 95% intervals; the grey band is the smallest difference of interest, ± 0.02. (A) *MYH7* under three benign sets (curated 12, any ClinVar 20, with population stand-ins 22). (B) S1 beyond the neighbour count in the ten genes (T2).

(the PM1 rule), so the circularity of the score is not removed by any analysis here. All benign sets are frequency-defined. The *MYH7* AUROC of S1 depends on the kernel radius (0.547 at 6 Å, 0.760 at 10 Å, 0.836 at 15 Å); the radius was fixed in advance and not tuned, and other radii were not tested in other genes. Simulated power is optimistic for the primary null, and the ten-point meta-regressions are exploratory. MyVariant.info serves a 2025 ClinVar snapshot older than our label file. The AlphaFold model is one static structure. The ClinGen cardiomyopathy specification of 2024 was not retrieved, so we used the values of Kelly et al. (2018). We did not compare against the spatial scans of Homburger et al. (2016) or Cadisch et al. (2025, preprint) because their patient-level data were not available, and no public functional benchmark overlaps our *MYH7* labels. The results are not for clinical use. Next steps are a repeat of the retrieval audit with a second annotation service and a benchmark that does not rest on clinical labels (Livesey and Marsh, 2025).

Conclusion

A single gene-symbol query to MyVariant.info lost more than 5% of ClinVar missense variants in every gene we tested, mainly through a lagging snapshot (about 70%) and partly through a symbol field that holds one symbol per record (about 30%). A simple count against the ClinVar release detects both. In a ten-gene panel chosen by a rule written before the cross-gene results and amended once, the shrunk 3D score S1 did not beat a neighbour count or AlphaMissense and added no measurable gain in nested models, although those equivalence results are weak.

Data and code availability

Code, derived tables, logs, the analysis plan with its SHA-256 hashes, and a checksum list of every archived download are at `https://github.com/aaryavvatts-lab/myh7-clinvar-retrieval-audit`. One script rebuilds all tables, figures and the paper from the archived downloads. All inputs are public except one frozen intermediate table from the earlier internal analysis, used only in one supplementary comparison. Licences of dbNSFP, AlphaMissense and other tool outputs must be checked before per-variant scores are redistributed.

Author contributions, funding and competing interests

A.S. is the sole author and is responsible for the content. The author received no funding and declares no competing interests.

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